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Top Skills

Aws ai
Responsible AI
Amazon Bedrock

Languages

English
Hindi

Certifications

Amazon Web Services Solutions
Architect - Associate
AWS AI practitioner Certification
Oracle Certified Professional, Java
SE 6 Programmer

Honors-Awards

Excellence Performance awards
Q4-2012
Spot Appreciation 2013
Spot Appreciation
Emirates Appreciation Najm
DC-AI Hackthon

Bharat Singh

Principal Engineer | Cloud-Native Platforms | AWS Certified (3x) |
Kubernetes | Java & Python | GenAI & Agentic Systems
Bengaluru, Karnataka, India

Summary

I'm a Principal Java and AWS Cloud Engineer with 14+ years of experience building resilient, scalable, and intelligent platforms across banking, aviation, healthcare, and government domains.

I specialize in designing cloud-native microservices and data-intensive systems that operate reliably at enterprise scale. My work focuses on transforming complex regulatory and business challenges into secure, cost-optimized, and high-performance architectures.

Currently, I lead architecture and hands-on development of large-scale stress testing and risk analytics platforms on AWS, leveraging Kubernetes, event-driven design, and distributed data processing to support real-world regulatory scenarios.

I remain strongly hands-on — actively coding, reviewing, and optimizing systems using Java, Python, Spring Boot, and modern cloud-native tooling. I enjoy solving hard performance and scalability problems and mentoring teams to build production-grade systems.

Alongside platform engineering, I'm actively exploring the future of software development through Generative AI and Agentic architectures. My recent work includes LLM-driven automation, RAG-based systems, and AI-assisted development workflows — including practical adoption of GitHub Copilot in enterprise engineering environments (which I'm also covering in my upcoming YouTube content).

I'm passionate about building systems that are not only scalable and reliable but also intelligent and future-ready.

Core Expertise

Cloud-Native Architecture • Microservices • AWS Platform Engineering • Kubernetes • Event-Driven Systems • Distributed Processing • Observability • Performance Optimization • Generative AI Integration

Experience

Standard Chartered

Lead Development

July 2023 - Present (3 years)

Singapore

- Designed and developed enterprise-grade stress testing models, improving system resilience and scalability while supporting regulatory and risk simulation workloads across multiple environments.
- Implemented scalable AWS cloud solutions using EKS, ECS, and AWS Lambda, enabling platforms to scale 3× during peak workloads while maintaining high availability.
- Developed high-performance applications and automation frameworks using Java, Python, and Kotlin, reducing manual operational effort by 30–40%.
- Architected and deployed highly available AWS infrastructure supporting mission-critical banking workloads, achieving 99.9%+ service availability.
- Streamlined CI/CD pipelines by enabling one-click Kubernetes Helm deployments, reducing deployment time by 60% and lowering release-related incidents.
- Designed and managed Kubernetes deployments, services, and batch jobs, optimising resource utilisation and reducing infrastructure costs by 20–25%.
- Improved code quality and security posture by proactively addressing vulnerabilities and enforcing best practices, resulting in fewer production defects and audit findings.
- Currently designing Agentic AI solutions leveraging LLMs, OpenAI, and agentic frameworks, accelerating intelligent automation initiatives and reducing analysis turnaround time by up to 50%.
- Designed and implemented AI agents using LangGraph for regulatory teams, enabling automated execution of complex stress scenarios and reducing scenario run time from hours to minutes.
- Contributed to the Catalyst Program, delivering innovation-led proof-of-concepts adopted by internal stakeholders and leadership teams.

Hitachi Vantara

Senior Consultant 2

August 2021 - July 2023 (2 years)

Bangalore Urban, Karnataka, India

- Led application modernization and cloud migration initiatives, focusing on scalable, secure, and highly available solutions.
- Architected and designed enterprise-grade applications and microservices using Java, Python, and Docker.
- Designed cloud-native architectures on AWS, leveraging managed services to improve reliability and performance.
- Enforced high code quality through best practices, design standards, code reviews, and automated testing.
- Collaborated with cross-functional teams to deliver production-ready solutions aligned with business and compliance requirements.

Projects

SurePoint – Single Sign-On (SSO) Platform

- Designed and developed a secure Single Sign-On (SSO) solution to centralize authentication and authorization for the SurePoint platform.
- Integrated PingIdentity for identity federation, authentication, and access management.
- Built backend services using Spring Boot, ensuring secure token handling and session management.
- Provisioned and managed cloud infrastructure using Terraform, enabling consistent and repeatable deployments.
- Improved user experience and security by reducing authentication complexity and strengthening access controls.

BCQMx – BeyondCare Quality Monitor (Sysmex | Healthcare Domain)

- Contributed to the development of an evidence-based quality control management system for monitoring analyzer health and accuracy in the healthcare domain.
- Designed and implemented microservices-based architecture to process and analyze large volumes of device and quality data.
- Leveraged AWS services such as ECS, Lambda, Kinesis, and SQS for scalable data ingestion, processing, and asynchronous communication.
- Built backend services using Python, ensuring performance, reliability, and compliance with healthcare standards.
- Utilized PostgreSQL for persistent storage and analytics, optimizing queries for performance and scalability.

- Implemented secure cloud networking and content delivery using VPC and CloudFront.

Robosoft Technologies

Technical Lead

May 2020 - August 2021 (1 year 4 months)

Bengaluru, Karnataka, India

Flipkart (Robosoft Payroll)

Domain: E-commerce

Project: Armstrong – Subscription Management Platform

Technologies: Java, Groovy, Python, Spring Boot, Dropwizard, Google Guice, MySQL, AWS, Kubernetes, Docker, Helm, CI/CD

Project Overview

Armstrong is a scalable subscription management platform that enables Flipkart users to purchase bundled subscriptions providing access to multiple partner applications at discounted prices. The system manages the complete subscription lifecycle, including onboarding, billing, automated renewals, upgrades, and cancellations, and is designed to support high traffic and large transaction volumes.

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Roles & Responsibilities

- Designed, developed, and maintained backend microservices for a large-scale subscription management system serving high-volume e-commerce traffic.
- Implemented end-to-end subscription lifecycle management, including bundled subscriptions, automated renewals, pricing, discounts, and entitlement management.
- Built scalable REST APIs using Java, Spring Boot, Dropwizard, and Google Guice, following microservices and domain-driven design principles.
- Designed and optimized MySQL data models and queries to handle millions of subscription records and transactions efficiently.
- Implemented fault-tolerant and idempotent workflows to ensure reliable billing and renewal processing.

- Containerized services using Docker and deployed them on AWS-hosted Kubernetes clusters, supporting horizontal scalability and high availability.
- Authored and maintained Helm charts and contributed to CI/CD pipelines for automated build, test, and deployment across environments.
- Monitored and tuned system performance to support thousands of concurrent users and high request throughput during peak traffic periods.
- Collaborated with product, QA, DevOps, and business teams to deliver production-ready solutions with strict reliability and SLA requirements.

TATWEER MIDDLE EAST AND AFRICA L.L.C

Full Stack Developer

June 2019 - March 2020 (10 months)

Abu Dhabi

Roles & Responsibilities

- Collaborated with clients and stakeholders to define project scope, functional requirements, and technical specifications.
- Designed and developed scalable, high-performance solutions across web, GIS, and smart system platforms.
- Built and optimized machine learning and computer vision models, including solutions for distracted driver detection.
- Developed full-stack web applications using Java and Angular, integrating backend services and data layers.
- Implemented GIS features such as ArcGIS mapping, spatial data visualization, and geofencing capabilities.
- Conducted machine learning experiments, model evaluation, and performance tuning for real-world deployment.
- Worked with cross-functional teams to deliver secure, reliable solutions for government and enterprise clients.

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Projects

IPV – Intelligent Patrolling Vehicle (Abu Dhabi Police)

Technologies: Python, Computer Vision, Machine Learning, Java, MongoDB

- Contributed to the development of an intelligent patrolling system for Abu Dhabi Police.
- Implemented machine learning and computer vision models for distracted driver detection and smart monitoring.

- Processed and analyzed video and sensor data to enable real-time detection and alerting.
- Integrated backend services and data storage using MongoDB to support analytics and reporting.

AIR – Asset Intelligence & Reporting (ArcGIS-based Asset Management)

Technologies: ArcGIS, GIS, Java, Angular, MongoDB

- Developed an ArcGIS-based asset management solution for tracking and monitoring assets.
- Implemented interactive maps, geospatial visualizations, and geofencing to support operational decision-making.
- Integrated GIS services with web applications using Java and Angular for real-time asset visibility.
- Enhanced system performance and usability through optimized spatial queries and data handling.

Technology Stack

Java, Angular, Python, ArcGIS, GIS, MongoDB, Machine Learning, Computer Vision, REST APIs

Emirates Group

4 years 6 months

Data Scientist

October 2017 - June 2019 (1 year 9 months)

Dubai, United Arab Emirates

Roles & Responsibilities

- Led end-to-end development of machine learning solutions, including problem definition, data collection, feature engineering, model training, evaluation, and deployment.
- Collaborated with business stakeholders to translate safety and revenue management requirements into data-driven AI/ML solutions.
- Designed and implemented NLP pipelines for text classification, prioritization, and automated decision support.
- Built, trained, and optimized machine learning models using Python, Scikit-learn, and TensorFlow to improve prediction accuracy and operational outcomes.
- Performed data preprocessing, exploratory data analysis (EDA), and feature engineering using Pandas and NumPy.

- Developed and deployed RESTful APIs using Flask for seamless integration with enterprise applications.
- Deployed, monitored, and maintained ML services on Azure Cloud, ensuring scalability, reliability, and security.
- Conducted model validation, performance tuning, and continuous improvement based on real-world feedback.
- Created technical documentation, model design documents, and deployment runbooks.

Project: Safeguard

Technologies: Python, NLP, Pandas, NumPy, TensorFlow, Flask, Azure Cloud

- Developed an NLP-based safety incident classification and prioritization system for global operations, automating triage and improving response times.
- Implemented ML models for severity-based prioritization, delivering an annual cost saving of ~AED 1.4 million.
- Deployed scalable REST APIs using Flask & Cloud.

Project: Group Materialization Forecasting

Technologies: Python, Scikit-learn, Pandas, NumPy, Flask

- Built a predictive ML model to forecast group booking materialization and optimize overbooking strategies.
- Analyzed historical booking and cancellation data to improve forecast accuracy and reduce revenue loss.
- Exposed prediction services via Flask APIs for enterprise system integration.

Full Stack Developer

January 2015 - September 2017 (2 years 9 months)

Dubai, United Arab Emirates

Roles & Responsibilities

- Contributed across the full software development lifecycle (SDLC), including requirement analysis, system design, development, testing, performance tuning, and deployment of enterprise web applications.
- Coordinated effectively between offshore and onsite teams to ensure timely delivery and alignment with business objectives.
- Led Proof of Concept (PoC) initiatives and delivered complex, scalable design solutions to address critical business requirements.
- Authored and maintained technical design documents, API specifications, and integration documentation.
- Modernized legacy applications by migrating to contemporary frameworks and architectures, improving performance, scalability, and maintainability.

- Managed build, release, and deployment processes for staging and QA environments, ensuring stable and repeatable deployments.

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Project: Hub Monitor

Technologies: Java, Spring Boot, Angular, Oracle, JUnit

- Designed and developed Hub Monitor, a decision support system for the Network Control Department.
- Built real-time operational dashboards providing situational awareness, forecasting, and performance insights.
- Enabled proactive issue identification and resolution, contributing to improved on-time aircraft performance and operational efficiency.
- Implemented backend services using Spring Boot and RESTful APIs, with Angular-based UI components for responsive user interaction.
- Ensured high code quality through unit testing using JUnit and adherence to best development practices.

Bureau Veritas Group

3 years 7 months

Senior Software Engineer

January 2013 - December 2014 (2 years)

Hong Kong, Hong Kong SAR

Roles & Responsibilities

- Collaborated with business stakeholders to gather, analyze, and clarify functional and non-functional requirements; performed impact analysis and developed Proof of Concepts (POCs) to validate technical feasibility.
- Designed and implemented reusable, scalable solutions and common frameworks adopted across multiple projects.
- Led and executed application modernization initiatives, including migration from GlassFish to JBoss and technology stack upgrades from JSF to HTML5, jQuery, and Spring Framework.
- Provided technical leadership by independently resolving complex system design challenges and ensuring adherence to best practices and architectural standards.
- Actively participated in team building by supporting technical interviews, onboarding, and mentoring junior developers.
- Designed and developed enterprise web applications using Java, Spring Boot, Hibernate, AngularJS, JSF, AJAX, and Oracle Database.

Software Engineer

June 2011 - December 2012 (1 year 7 months)

Noida Area, India

Roles & Responsibilities

- Involved in end-to-end software development lifecycle including requirement gathering, analysis, design, development, deployment, and maintenance.
- Worked on projects from scratch, contributing to architecture and core feature development.
- Participated in client discussions to understand requirements and provide technical solutions.
- Maintained mandatory project documentation and process compliance artifacts.
- Implemented new solutions and enhancements based on evolving business needs.
- Performed unit testing and integration testing to ensure application quality and reliability.

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Applications

1) BVLABNG (Bureau Veritas Lab NG)

- Developed an application to automate laboratory testing processes for Bureau Veritas.
- The system manages and stores product testing results and related laboratory data, improving efficiency and traceability of test operations.

2) CDM (Client Directory Management)

- Worked on a centralized Client Management System for CPS.
- Implemented support for multilingual data storage, including Chinese (Simplified), Korean, and Vietnamese languages.

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Tools & Technologies

- Languages & Frameworks: Java, Servlets, JSP, JSF 2.0
- Frontend: jQuery, JavaScript
- Database: Oracle
- Application Server: GlassFish v2
- Build & Tools: Maven, NetBeans

- Data Access: JDBC

RoseIndia Technology

Software Engineer

July 2010 - May 2011 (11 months)

New Delhi, India

Roles & Responsibilities

- . Gathered and documented requirements for new solutions in alignment with business needs.
- Analyzed business problems, resolved technical issues, and developed application modules.
- Participated in unit testing and integration testing to ensure quality and stability.
- Responsible for bug fixing, application support, and ongoing project maintenance.
- Performed system design, analysis, and end-to-end application development.
- Designed logical and physical data models; generated DDL scripts and wrote DML scripts for Oracle 10g databases.

Education

Dr. A.P.J. Abdul Kalam Technical University

Master of Computer Applications - MCA, Computer Science · (2007 - 2010)

Kumaun University

Bachelor of Science - BS · (2004 - 2007)

Udacity

Machine Learning NanoDegree, Machine learning